

Week 4 Assignment: Planning in a Stochastic Gridworld

MDP modeling, value iteration, policy iteration, and verification

CMPSC 480 - Student Question Deck

Assessment overview

Model stochastic transitions precisely and solve the same MDP with two dynamic-programming methods.

Coverage: Week 4: MDPs, Bellman backups, value iteration, and policy iteration

Points

10 total

Expected time

6-8 hours

Submission

Repository plus experiment memo

Learning objectives

- ▶ Encode a finite MDP with normalized transition distributions.
- ▶ Implement terminal-aware Bellman backups.
- ▶ Run value iteration to a documented tolerance.
- ▶ Alternate policy evaluation and improvement correctly.
- ▶ Cross-check two solvers and reason about edge cases.

Requirements checklist

- ▶ Use Python 3.11 and NumPy only unless the prompt explicitly permits another package.
- ▶ Preserve the published interfaces and make all tie-breaking deterministic.
- ▶ State assumptions, validate inputs, and handle empty, terminal, and unreachable cases.
- ▶ Include focused tests whose names explain the defect each test would expose.
- ▶ Report measured evidence; do not replace experimental results with unsupported claims.

Task 1 - MDP model (2 points)

Represent grid states, walls, terminal rewards, legal actions, and 80/10/10 stochastic motion.

- a. Define S, A, P, R, gamma, initial state, walls, and terminal states. [1 points]
- b. Write validation tests for every state-action transition distribution. [1 points]

Task 2 - Value iteration (3 points)

Implement synchronous Bellman optimality backups and a convergence test.

- a. Write the Bellman optimality backup used for each nonterminal state. [1 points]
- b. Use a copy of the previous value vector and stop when the maximum residual is below tolerance. [1 points]
- c. Extract a deterministic greedy policy and report values for designated cells. [1 points]

Task 3 - Policy iteration (3 points)

Alternate policy evaluation with greedy policy improvement.

- a. Describe iterative policy evaluation for a fixed policy. [1 points]
- b. Implement greedy improvement and state the stability condition. [1 points]
- c. Compare the final policy and values with value iteration. [1 points]

Task 4 - Validation and analysis (2 points)

Use analytic and metamorphic checks rather than trusting one implementation.

- a. Test $\gamma=0$, a one-state terminal MDP, blocked slips, and a reward perturbation. [1 points]
- b. Explain how γ and slip probability change the policy in two experiments. [1 points]

Edge cases and defensive programming

- ▶ Initial state is terminal.
- ▶ A slip hits a wall or boundary.
- ▶ Two actions have equal value.
- ▶ Gamma equals zero.
- ▶ Tolerance is nonpositive.

Submission instructions

- ▶ Submit source, tests, and exact run commands.
- ▶ Include residual and policy-comparison tables.
- ▶ State reward and terminal conventions in the README.

Do not submit credentials, generated caches, or unapproved libraries.

Grading rubric

Category	Points	Required evidence
MDP model	2	Validated transition and reward functions.
Value iteration	3	Value iteration, policy extraction, and residual trace.
Policy iteration	3	Policy iteration and comparison with value iteration.
Validation and analysis	2	Tests and a concise result table.

Reflection prompt

In 150-200 words:

Describe one modeling choice whose effect was larger than an algorithmic optimization.
Support the claim with a controlled comparison.